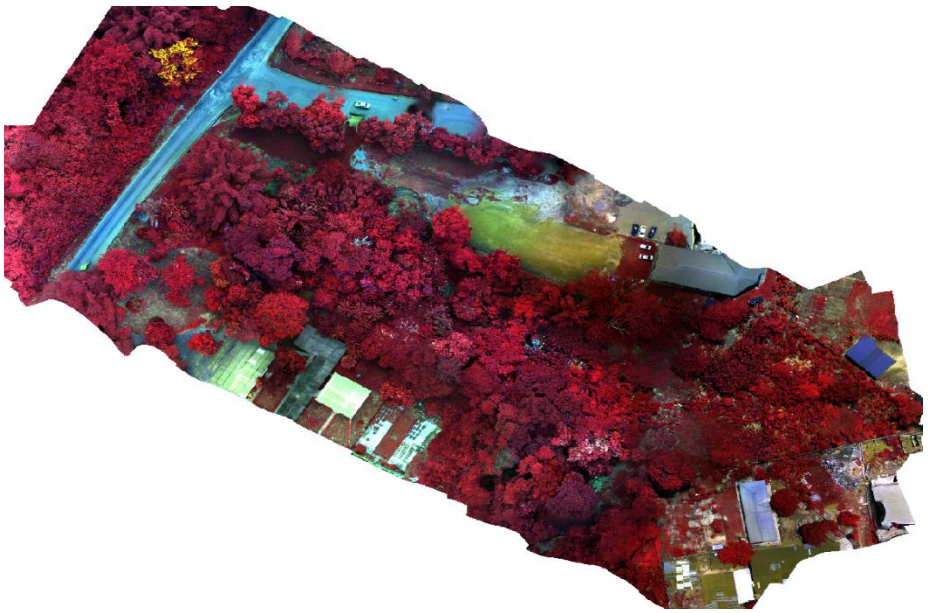
RESEARCH PORTFOLIO

Abigail A. Adjetey Bisa

Remote Sensing | *GIS* | *Artificial Intelligence* | *Agricultural & Environmental Applications*

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Research Profile

My research interests lie at the intersection of remote sensing, geographic information science, and artificial intelligence for agricultural and environmental applications. My experience includes UAV multispectral remote sensing for crop nutrient assessment, satellite-based land-use/land-cover classification, deep learning for satellite image classification, and GIS-based analysis of cocoa agroforestry landscapes. Across these projects, I have worked with field and reference data, satellite and UAV imagery, spatial data processing, machine-learning and deep-learning methods, and the production of spatially explicit outputs for environmental and agricultural decision-making.

Research Themes

Remote Sensing
Satellite and UAV multispectral imagery, image processing, vegetation indices, and time-series analysis.

Geospatial AI
Machine learning and deep learning for classification, prediction, and spatially explicit environmental applications.

Agricultural Applications
Crop nutrition, cocoa agroforestry, precision agriculture, and landscape monitoring.

Environmental Applications
Deforestation monitoring, land use/land cover mapping, and sustainable land management.

Research Trajectory

My research trajectory has progressively developed from field-based environmental data collection and GIS analysis toward increasingly integrated Earth observation and AI workflows. This progression has provided experience across the full research chain: field data collection → remote sensing → geospatial processing → machine learning/deep learning → spatial prediction and decision-support products.

Selected Research Experience

Research project	Core methods	My contribution
Cocoa nutrient assessment (MSc)	UAV multispectral imagery; vegetation indices; machine learning	Developed and evaluated predictive models for cocoa foliar N, P and K using UAV-derived spectral information.
Deforestation-Free Cocoa	Satellite time series; CNN/deep learning; ML classification; Python	Led/participated in field training-data collection and developed satellite classification models for historical landscape mapping.
Shade characterisation in cocoa agroforestry	Sentinel-2; spectral/texture/ biophysical predictors; Random Forest	Worked on the remote-sensing and geospatial component used to classify cocoa farms by shade level.
High Forest Zone LULC work	Satellite remote sensing; GIS; reference/training data; classification	Executed the LULC/remote-sensing work assigned by my supervisor within the broader research project.

Attribution principle: the portfolio distinguishes the broader project from my individual contribution. Work performed by supervisors, co-authors, or other team members is not presented as my own.

Research Skills Demonstrated Across Projects

- Earth observation data acquisition, preprocessing and analysis
- UAV multispectral data processing and interpretation
- Satellite image time-series analysis
- Machine learning and deep learning for geospatial data
- Raster and vector GIS analysis
- Spatial interpolation and geostatistical analysis
- Field data collection and ground/reference data generation
- Scientific writing, research communication and publication

PROJECT 1

UAV Multispectral Remote Sensing and Machine Learning for Cocoa Nutrient Assessment

MSc research | Juaboso cocoa landscape, Western North, Ghana

Research Context

Cocoa production in Ghana faces challenges associated with declining soil fertility and the limitations of blanket fertilizer recommendations. My MSc research investigated the use of UAV-based multispectral remote sensing and machine-learning approaches to assess cocoa foliar nitrogen (N), phosphorus (P), and potassium (K), while also examining relationships between soil and leaf nutrient concentrations. This research was one component of a broader project, **FRAME-Cocoa Project**, sponsored by African Plant Nutrition Institute (APNI).

Research Focus

The thesis had two linked objectives: (1) to quantify relationships between soil and leaf N, P and K concentrations; and (2) to develop predictive models for cocoa leaf nutrient concentrations using spectral bands and derived vegetation indices. The study was positioned as a contribution to precision agriculture and spatially informed nutrient assessment.

Research Question

Can UAV multispectral imagery and machine-learning models be used to estimate cocoa leaf nitrogen (N), phosphorus (P), and potassium (K) concentrations and produce continuous spatial predictions across a cocoa production landscape?

Data and Remote Sensing Platform

- DJI Matrice 350 RTK UAV, operated at 150 m altitude and 3 m/s with 65% image overlap.
- MicaSense RedEdge-P multispectral sensor with Blue (475 nm), Green (560 nm), Red (668 nm), Red Edge (717 nm), and NIR (842 nm) bands.
- Blue, Green, Red, Red Edge and NIR bands and derived vegetation indices.
- 59 composite soil samples/field sampling locations.
- Ground-truth soil and foliar N, P and K measurements used for model development and evaluation.

PROJECT 1

UAV Multispectral Remote Sensing and Machine Learning for Cocoa Nutrient Assessment

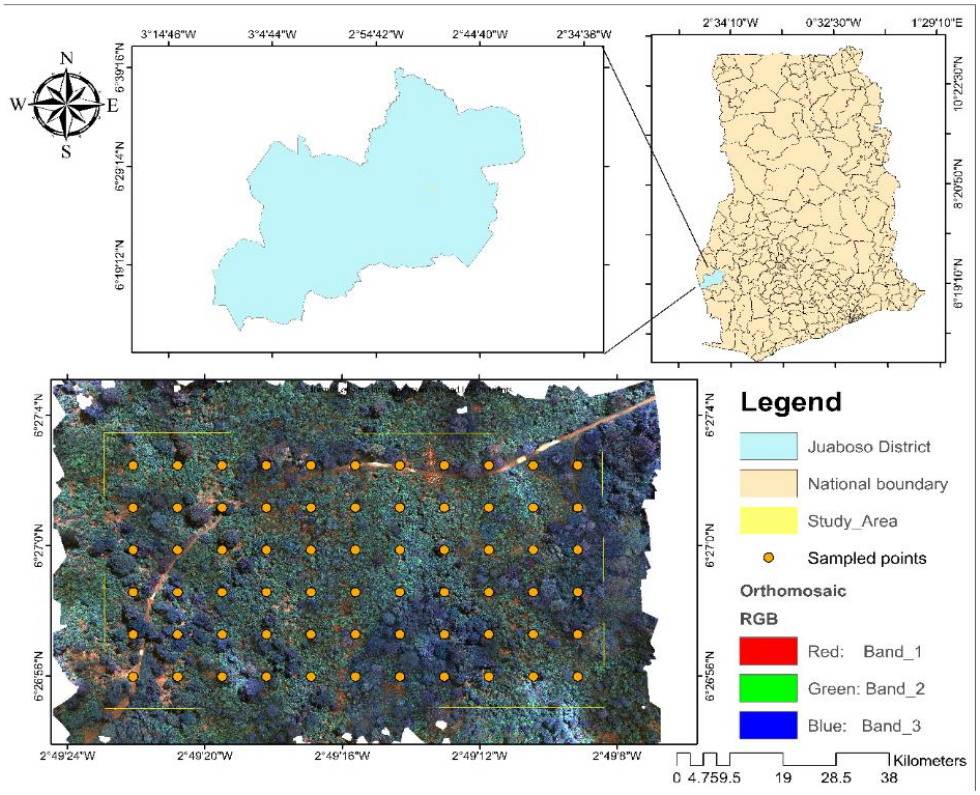


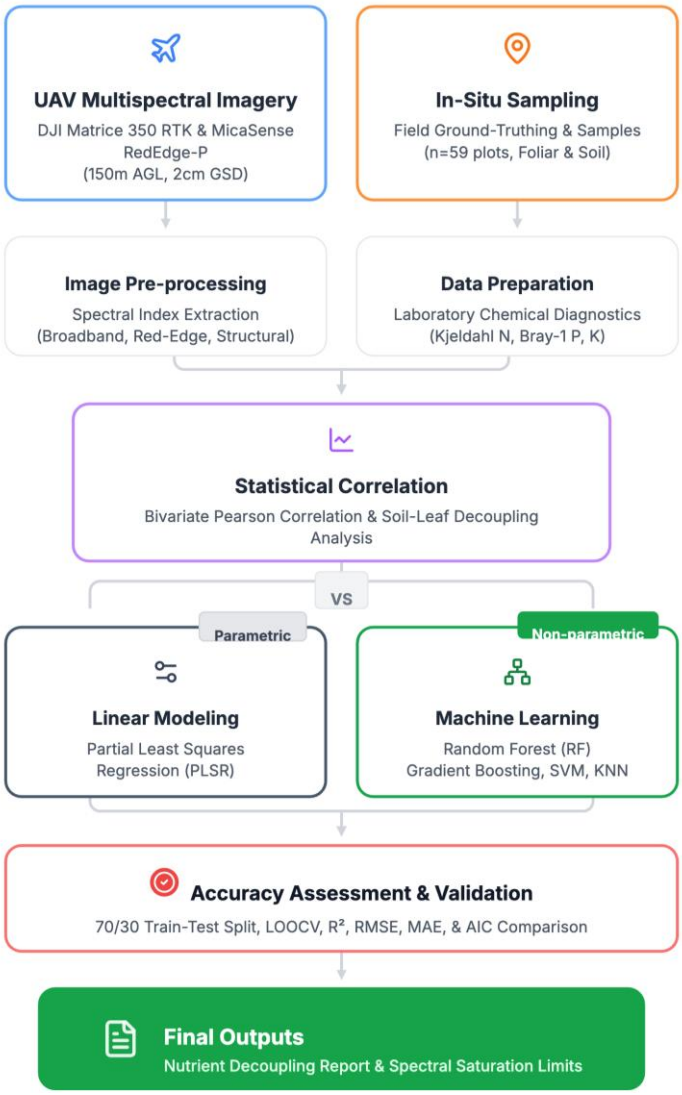
Figure 1. Map of study area displaying UAV orthomosaic of selected farm site under investigation.

PROJECT 1

UAV Multispectral Remote Sensing and Machine Learning for Cocoa Nutrient Assessment

Methods and modelling

Analytical Workflow



Remote Sensing and Data Processing

- Imagery was radiometrically corrected using the MicaSense calibration panel and processed in Agisoft Metashape to generate a 0.09 m GSD orthomosaic.
- Spectral bands and vegetation indices were derived to represent canopy spectral behaviour, chlorophyll-related properties and vegetation structure.
- Raster predictors were prepared and extracted at the field sampling locations for subsequent statistical and machine-learning analysis.
- Vegetation-index predictors with inter-correlation greater than 0.95 were excluded to reduce redundancy and multicollinearity.
- The resulting spectral predictor set was used to compare machine-learning models for foliar nutrient estimation.

Machine-Learning Models

- Random Forest (RF)
- Support Vector Regression (SVM)
- Artificial Neural Network (ANN)
- Partial Least Squares Regression (PLSR)
- Extreme Gradient Boosting (XGBoost)
- Gradient Boosting Machine (GBM)

Model Selection

Target	**Key result in final submitted thesis**
Leaf N	Spectral models: very low R^2; soil-based RF $R^2 = 0.17$
Leaf P	Spectral models: XGBoost highest $R^2 \approx 0.09$; soil-based RF $R^2 = 0.55$
Leaf K	Spectral models: ANN $R^2 \approx 0.10$; soil-based RF $R^2 = 0.52$

The submitted thesis reports nutrient-specific and model-specific results, with Random Forest generally performing strongly in the soil-to-leaf analysis and XGBoost showing the strongest spectral-variable result for Leaf P.

PROJECT 1

UAV Multispectral Remote Sensing and Machine Learning for Cocoa Nutrient Assessment

Results and research contribution

Research Output

The modelling results demonstrated that relationships between UAV-derived spectral information and cocoa foliar nutrients varied by nutrient. Different algorithms therefore provided varying predictive performance for nitrogen, phosphorus and potassium. Phosphorus and potassium showed stronger predictive signals than nitrogen, while limited nutrient variability constrained the performance of spectral models. In the soil-to-leaf analysis, Random Forest produced the strongest fits for Leaf N, Leaf P and Leaf K (R^2 = 0.17, 0.55 and 0.52, respectively). For spectral-variable models, performance was generally weak: nitrogen predictions were close to zero R^2, XGBoost performed best for Leaf P ($R^2 \approx 0.09$), and ANN had the highest R^2 for Leaf K (≈ 0.10), although with relatively high errors.

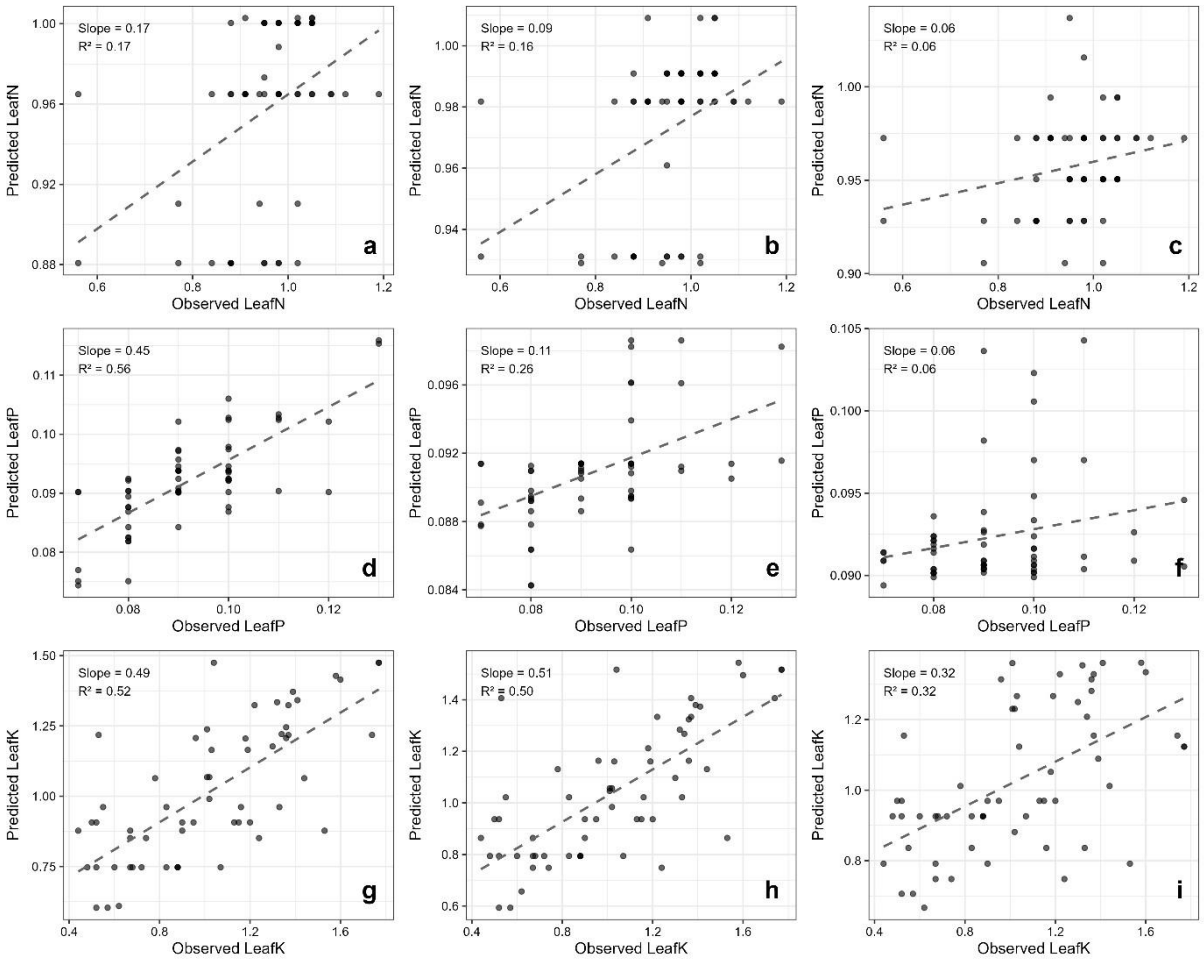


Figure 2. Comparison of observed vs predicted foliar nutrient concentrations (LeafN, LeafP, LeafK) using three machine learning models: Random Forest (plots a, d, g), Support Vector Machine (plots b, e, h), and Artificial Neural Network (plots c, f, i).

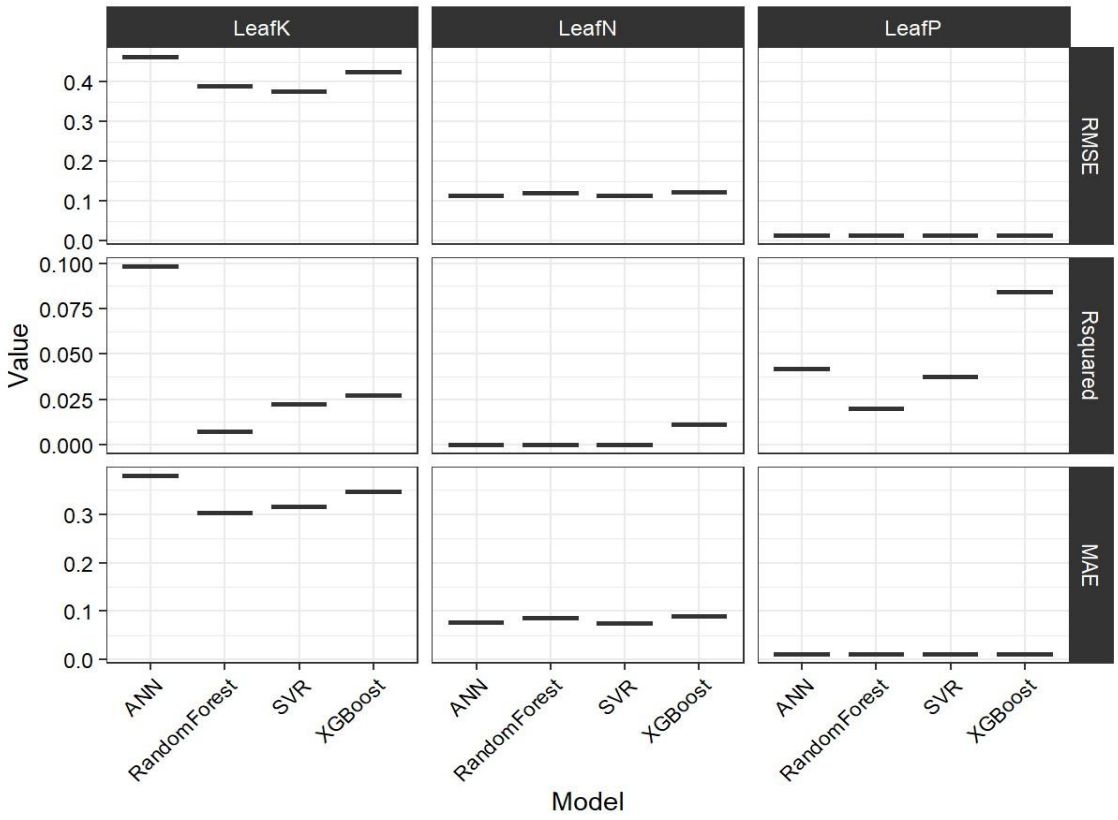


Figure 3. Comparison of model performance in predicting leaf nutrient concentrations using different algorithms (RF, XGBoost, SVM, and ANN) based on R- squared values, RMSE scores and MAE

Research Contribution

My contribution covered the UAV and field-data components, multispectral predictor preparation, feature screening, statistical analysis of soil–leaf nutrient relationships, comparative machine-learning modelling, and interpretation of spectral predictors and model behaviour. The work demonstrates how high-resolution Earth observation and AI can support more spatially informed approaches to crop nutrient assessment and management.

PROJECT 2

Geospatial Characterisation of Shade Canopy in Cocoa Agroforestry and Soil Resources

Juaboso District, Ghana | Remote-sensing component under FRAME-Cocoa Project

Research Context

This is co-authored research which included field-based and GIS-driven characterisation of shade canopy in cocoa agroforestry farms within one of Ghana's prime cocoa landscapes. Shade canopy in cocoa farms can yield its highest benefits when managed in an optimum range, however, can also be detrimental to cocoa productivity when it falls short or goes beyond that specific range. This work integrated remote sensing, field observations and mathematical optimisation.

Data and Field Research

- Collected soil samples at multiple depths from cocoa farms across Juaboso District.
- Identified and measured canopy sizes of shade trees present within sampled farms.
- Conducted and supported laboratory analysis of soil properties.
- Characterised shade-tree cover and cocoa agroforestry structure.

My Remote-Sensing Contribution

In addition to the field data collection, I worked on the use of Sentinel-2-derived information to characterise cocoa farms according to shade level. The remote-sensing workflow used spectral, texture, and biophysical predictors and Random Forest modelling.

Remote-Sensing Predictors

- Sentinel-2 imagery covering dry season observations.
- Spectral variables from Sentinel bands and vegetation indices.
- Texture variables derived using GLCM metrics.
- Biophysical variables including LAI, fAPAR, fCover, chlorophyll content and canopy water content.

Classification Approach

Random Forest modelling was used to predict continuous canopy-cover information, which was subsequently categorised into four shade levels; no shade, low shade, moderate shade (optimal), and high shade.

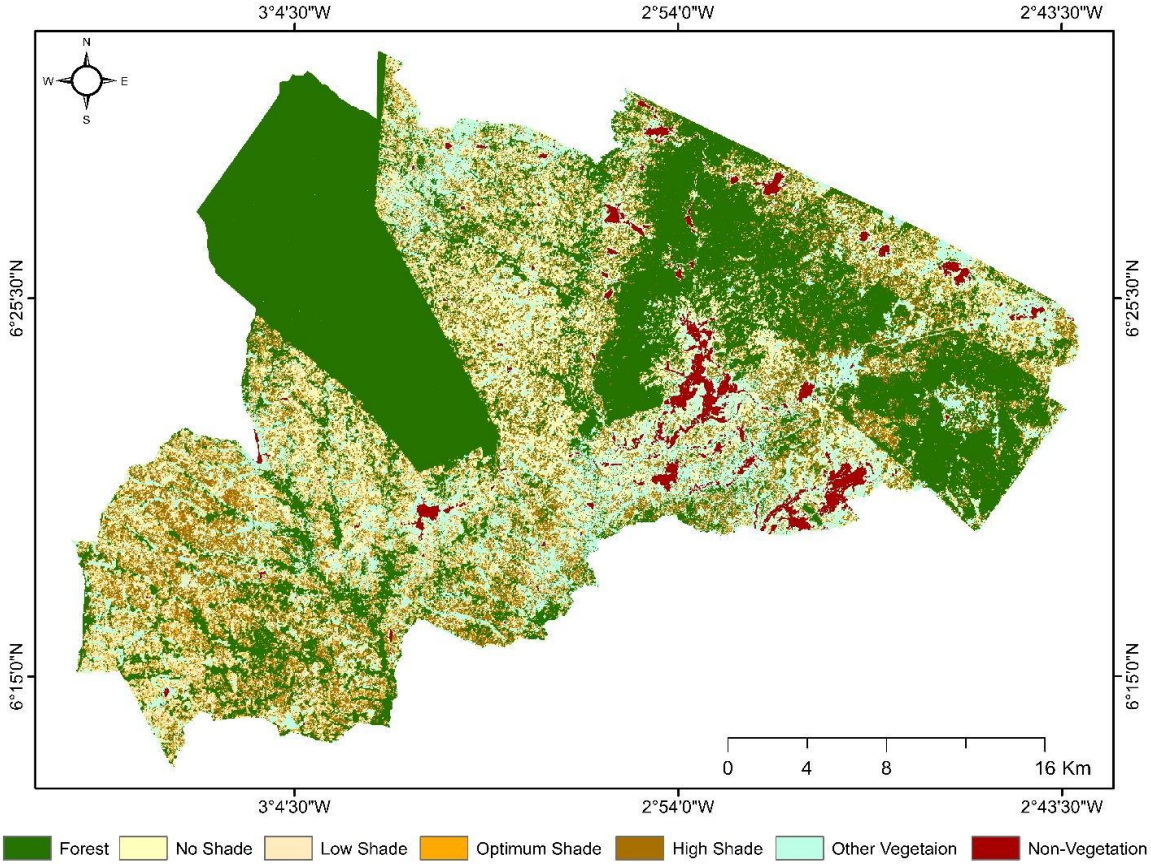


Figure 4. Land Cover Map of the Juaboso Landscape including shade canopy variations of cocoa agroforestry farms within the landscape.

Model Configuration

- Random Forest was used for supervised prediction of continuous canopy cover.
- Backwards feature elimination was applied to identify relevant predictors and reduce redundancy.
- Model performance was assessed using 10-fold cross-validation.

My Contribution vs. Broader Study

My contribution	Broader study
Remote sensing / GIS processing and shade-level classification	Integration of field tree data and shade-tree species information
Sentinel-derived predictor development and Random Forest analysis	Mathematical optimisation and site-specific tree-combination recommendations
Spatial characterisation of cocoa shade conditions	Overall agroforestry management framework

PROJECT 3

Satellite Time-Series Classification for Deforestation-Free Cocoa Monitoring

Deforestation-Free Cocoa Project | Ghana

Research Context

Within the Deforestation-Free Cocoa Project, I worked on the remote-sensing and GIS components supporting historical land-use/land-cover mapping across cocoa landscape hotspot intervention areas: ***Sefwi-Wiawso Bibiani***, ***Asunafo-Asutifi*** and ***Juaboso-Bia***. The project provided practical experience in using temporal Earth observation information rather than relying only on single-date imagery.

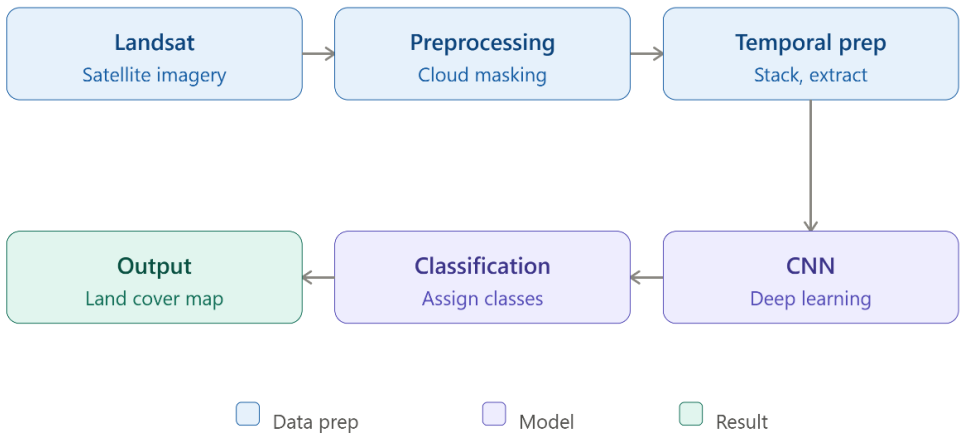
My Role

- Led a team involved in field data collection for the mapping work.
- Supervised collection of training coordinates for historical years 2015, 2020 and 2024.
- Developed deep-learning and machine-learning models for satellite image classification using Python.
- Supported production of historical LULC information for the intervention landscapes.
- Constructed physical 3D landscape models for stakeholder communication.

CNN Component

A key part of my technical experience was applying a Convolutional Neural Network (CNN) to satellite image time-series classification. This provided direct experience applying deep learning to multi-temporal Earth observation data.

Deep-Learning Workflow



PROJECT 3

Satellite Time-Series Classification for Deforestation-Free Cocoa Monitoring

Outputs and significance

Training and Reference Data

The project involved systematic collection and preparation of field training/reference information to support historical classification. The project documentation records spatially distributed training-data collection and development of historical LULC maps.

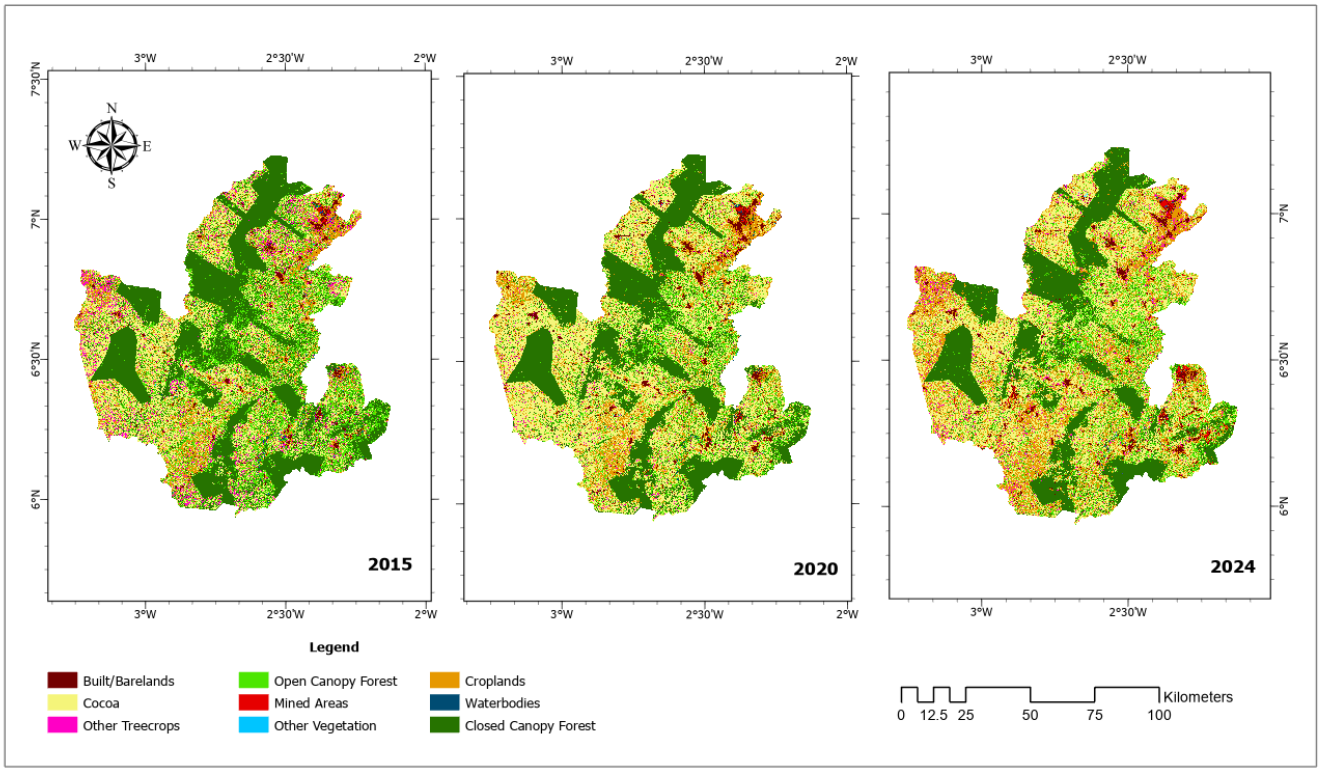


Figure 5. Time-series map displaying changes in landscape from 2015 to 2024.

Research Significance

The project expanded my experience from UAV-based crop assessment to satellite-scale environmental monitoring and from conventional machine learning to deep learning. It also strengthened my ability to connect field/reference data with multi-temporal satellite classification and equipped me with the skill of transforming maps into 3D landscape models.

PROJECT 4

Land-Use/Land-Cover Classification in Ghana's High Forest Zone

Remote sensing, GIS and reference-data based classification

Project Context

This work originated as part of broader research led by my supervisor on forest and land-cover mapping in Ghana's High Forest Zone. Under this research, we sought to investigate which algorithm best distinguishes land use classes in a highly heterogenous landscape such as the high forest zone of Ghana.

My Contribution

- Executed the assigned remote-sensing and GIS component.
- Prepared and analysed satellite imagery for land-cover classification.
- Worked with training/reference data for supervised classification.
- Produced and interpreted LULC classification outputs.
- Applied geospatial analysis to support the broader forest/landscape research.

Technical Context

The High Forest Zone research addresses forest mapping in a spectrally heterogeneous landscape and considers medium-resolution Landsat and Sentinel Earth observation datasets, including comparison of traditional machine learning classifiers and CNN-based deep learning.

LULC Workflow

- Satellite imagery preparation and preprocessing.
- Preparation/use of training and reference data.
- Supervised land-cover classification.
- Accuracy assessment and spatial interpretation.

PROJECT 4

Land-Use/Land-Cover Classification in Ghana's High Forest Zone

Classification Output

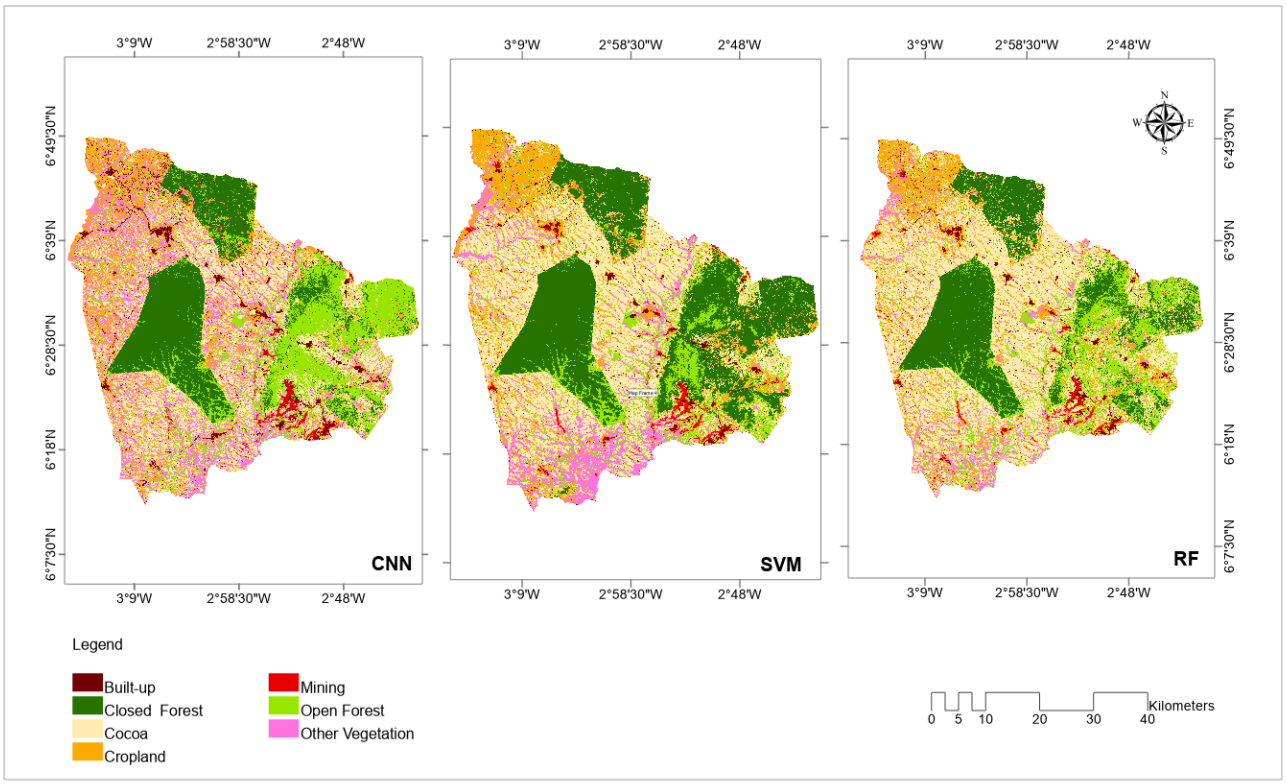


Figure 6. Comparative spatial distribution of land use and land cover (LULC) classification across the Juaboso-Bia HIA using Random Forest (RF), Support Vector Machine (SVM), and Convolutional Neural Network (CNN) classifiers on LANDSAT 9 L2 imagery

Publications & Research Outputs

Peer-Reviewed Publication

Ashiagbor, G., Essah, I.S., Baiden, P.P.K., **Adjetey Bisa, A.A.,** et al. (2026). Modelling the number of shade tree species needed to reach the optimal canopy target in cocoa agroforestry systems. *Agroforest Syst* **100**, 36 (2026).

Manuscripts Under Review

- Ashiagbor, G., Ahorlu, O., Enyinful-Andoh, G., **Adjetey Bisa, A. A.**, Essah, I. S., Tawiah, W. B., Yeboah, E., Mutegi, J., Mugi-Ngenga, E., Amouzou, K. A., & Oberthür, T. (2026). Shade Tree Structural Management Does Not Predict Soil Chemical Quality or Cocoa Yield in Smallholder Agroforestry: Plot-Level Evidence from Ghana. (Manuscript No. AGEE58523). — *Under Review at Agriculture, Ecosystems and Environment*

Manuscript in Preparation

Predicting Macronutrient Status in Mature Cocoa Using UAV Multispectral Imagery and Machine Learning — In Preparation

MSc Thesis

Predicting Macronutrient Levels in Cocoa Soils Using Multispectral Unmanned Aerial Vehicles (UAV) — Master of Science (MSc.) Geo-Information Science for Natural Resources Management, KNUST, 2025.

Selected Research Outputs

- UAV multispectral orthomosaics and derived vegetation-index products.
- Cocoa foliar nutrient modelling outputs.
- Satellite time-series classification outputs.
- Sentinel-based cocoa shade classification maps.
- Land-use/land-cover and landscape classification products.

Technical Capabilities

Earth Observation
Satellite remote sensing; UAV multispectral remote sensing; image preprocessing; vegetation indices; time-series analysis.

GIS & Spatial Analysis
ArcGIS Pro; QGIS; raster analysis; spatial analysis; kriging/interpolation; cartography.

Artificial Intelligence & Machine Learning
Convolutional Neural Networks (CNNs); Artificial Neural Networks (ANNs); Random Forest; XGBoost; Support Vector Machines; Gradient Boosting; PLSR.

Programming & Analysis
Python; R/RStudio; Google Colab.

Remote Sensing Software
SNAP; Agisoft Metashape.

UAV / Field Research
DJI Matrice 350 RTK; MicaSense RedEdge-P; field sampling; georeferenced observations; environmental and crop data collection.

Links

- Repository:https://drive.google.com/drive/folders/1oQW2PkMrRqszJHe6Tyt9hNYEWxkXkR0M?usp=drive_link
- Published paper(s): https://doi.org/10.1007/s10457-025-01402-6
- Some scientific writings/ reports: https://drive.google.com/drive/folders/1iTtz8tL972xv826KnH7IIDWEJrI1BXwJ?usp=drive_link